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# DSA Syllabus with JS

## Complete Coding Journey – From Basics to Placement Level Mastery

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- Near 250 recorded video solutions covering important problems
- 25+ coding patterns covered (problem-solving focused)
- Advanced DSA concepts included
- 300 MCQs for concept clarity and practice
- 100+ assignment questions for hands-on coding
- One live session every week
- Mentor support on Discord
- Interview guidance
- Handwritten notes
- Online coding challenges

### Bonus Sheets:

- Company-focused interview questions
- Service-based & product-based question sets

## BASICS OF PROGRAMMING

### Concepts Covered

- Variables, Data Types (let, var, const)
- Input/Output operations
- Operators (Arithmetic, Relational, Logical, Unary)
- Expressions & Formulas
- Math API (abs, ceil, floor, round, pow, sqrt, random, etc.)
- Type Coercion

### Problem Table

| # | Problem                             | Difficulty | Pattern                   |
|---|-------------------------------------|------------|---------------------------|
| 1 | Sum of two integers                 | Easy       | Input → Process → Output  |
| 2 | Relation between integer and string | Easy       | Type Coercion             |
| 3 | Sum and message                     | Easy       | Expression Evaluation     |
| 4 | Accept and print the answer         | Easy       | Input → Output Formatting |

| # | Problem                             | Difficulty | Pattern                          |
|---|-------------------------------------|------------|----------------------------------|
| 5 | Swap two variables (3 methods)      | Easy       | Swap Pattern                     |
| 6 | Calculate Compound Interest         | Easy       | Mathematical Simulation          |
| 7 | Generate OTP                        | Easy       | Math.random()                    |
| 8 | Area of triangle by Heron's formula | Medium     | Formula + Multi-step Calculation |
| 9 | Circumference of circle             | Easy       | Formula Based                    |

## CONTROL FLOW STATEMENTS

### Concepts Covered

- if, if-else
- Nested conditions
- Ternary operator
- Type casting

### Problem Table

| #  | Problem                            | Difficulty | Pattern                   |
|----|------------------------------------|------------|---------------------------|
| 10 | Greatest between two numbers       | Easy       | Comparison                |
| 11 | Check even or odd                  | Easy       | Modulo Check              |
| 12 | Voter eligibility check            | Medium     | Conditional + Calculation |
| 13 | Greatest among three numbers       | Easy       | Nested Comparison         |
| 14 | Leap year check                    | Easy       | Mathematical Condition    |
| 15 | Shop discount calculation          | Medium     | Conditional + Formula     |
| 16 | Electricity bill calculation       | Medium     | Slab Based Calculation    |
| 17 | Counting number of days in a month | Medium     | Conditional Mapping       |

## LOOPS & SWITCH STATEMENT

### Concepts Covered

- for, while, do-while
- Nested loops
- break, continue
- switch statement
- Pattern printing (nested loops)

## Problem Table

| #  | Problem                                      | Difficulty | Pattern                         |
|----|----------------------------------------------|------------|---------------------------------|
| 18 | Print 'Hello World' n times                  | Easy       | Iteration                       |
| 19 | Print natural numbers up to n                | Easy       | Iteration                       |
| 20 | Reverse for loop – Print n to 1              | Easy       | Reverse Iteration               |
| 21 | Multiplication table of a number             | Easy       | Iteration + Formula             |
| 22 | Sum up to n terms                            | Easy       | Accumulation                    |
| 23 | Factorial of a number                        | Easy       | Recursion (Multiplicative)      |
| 24 | Sum of even & odd numbers separately         | Easy       | Conditional + Accumulation      |
| 25 | Print all factors of a number                | Medium     | Factor Pattern                  |
| 26 | Prime number check                           | Medium     | Factor Pattern                  |
| 27 | Power of a number ( $a^b$ )                  | Medium     | Repeated Multiplication         |
| 28 | Sum of digits                                | Medium     | Digit Extraction + Accumulation |
| 29 | Reverse a number                             | Medium     | Digit Manipulation              |
| 30 | ISBN number check                            | Medium     | Mathematical Pattern            |
| 31 | Automorphic number                           | Medium     | Mathematical Pattern            |
| 32 | Check vowel/consonant using switch           | Easy       | Character Mapping (Switch)      |
| 33 | Area of circle, rectangle, triangle (switch) | Medium     | Menu / Switch                   |
| 34 | Repeat hello (do-while)                      | Easy       | Sentinel Loop                   |
| 35 | Guess the number                             | Medium     | Simulation + Loop               |
| 36 | Sasta Calculator (do-while)                  | Medium     | Menu Loop                       |
| 37 | Project – Restaurant menu                    | Medium     | Simulation + Loop               |

## Pattern Problems

## Problem Table

| #  | Problem                                 | Difficulty | Pattern             |
|----|-----------------------------------------|------------|---------------------|
| 38 | Right Angle Triangle – Star Pattern     | Easy       | Nested Loop Pattern |
| 39 | Right Angle Triangle – Number Pattern   | Easy       | Nested Loop Pattern |
| 40 | Right Angle Triangle – Alphabet Pattern | Easy       | Nested Loop Pattern |
| 41 | Inverted Right Angle Triangle Pattern   | Medium     | Nested Loop Pattern |
| 42 | Mirror Right Angle Triangle Pattern     | Medium     | Nested Loop Pattern |
| 43 | Triangle (Centered) Pattern             | Medium     | Nested Loop Pattern |

| #  | Problem   | Difficulty | Pattern       |
|----|-----------|------------|---------------|
| 44 | V Pattern | Medium     | Pattern Logic |
| 45 | X Pattern | Medium     | Pattern Logic |

## ARRAY DATA STRUCTURES (LEVEL – 1)

### Concepts Covered

- Arrays (1D), indexing
- Traversal
- Reversal, Rotation
- Basic array operations

### Problem Table

| #  | Problem                              | Difficulty | Pattern                  |
|----|--------------------------------------|------------|--------------------------|
| 46 | Sum & average of array elements      | Easy       | Traversal + Accumulation |
| 47 | Find greatest element                | Easy       | Max Tracking             |
| 48 | Find second greatest element         | Medium     | Max Tracking             |
| 49 | Reverse an array                     | Easy       | Traversal + Reverse      |
| 50 | Move negatives left, positives right | Medium     | Partition / Two Pointer  |
| 51 | Array left rotation by 1             | Easy       | Rotation                 |
| 52 | Array left rotation by K             | Medium     | Rotation                 |
| 53 | Count subarrays with given sum       | Medium     | Prefix Sum               |
| 54 | Linear Search                        | Easy       | Traversal                |
| 55 | Binary Search                        | Easy       | Binary Search            |

## SORTING ALGORITHM

### Concepts Covered

- Stable vs Unstable Sort
- Comparison-based sorting
- In-place sorting

### Problem Table

| #  | Problem        | Difficulty | Pattern               |
|----|----------------|------------|-----------------------|
| 56 | Bubble Sort    | Easy       | Comparison + Swapping |
| 57 | Insertion Sort | Easy       | Insertion + Shifting  |
| 58 | Selection Sort | Easy       | Selection + Swapping  |

## MATRIX (2D ARRAYS)

### Concepts Covered

- 2D Arrays
- Matrix traversal
- Index handling & exceptions

### Problem Table

| #  | Problem             | Difficulty | Pattern                |
|----|---------------------|------------|------------------------|
| 59 | Matrix Diagonal Sum | Easy       | Traversal              |
| 60 | Transpose Matrix    | Easy       | Index Transformation   |
| 61 | Flipping an Image   | Easy       | Two Pointer (Row-wise) |
| 62 | Rotate Image        | Medium     | Transpose + Reverse    |
| 63 | Spiral Matrix       | Medium     | Boundary Simulation    |
| 64 | Set Matrix Zeroes   | Medium     | Marking / In-place     |

## TIME AND SPACE COMPLEXITY

### Concepts Covered

- Time Complexity (Big-O)
- Space Complexity
- Complexity equations
- TLE (Time Limit Exceeded)
- Constraints & judge rules
- Handling for large input

## STRING

### Concepts Covered

- String literals, immutability

- String methods (slice, substring, toUpperCase, trim, indexOf, split, etc.)
- Character operations (charAt, charCodeAt)
- Traversal & manipulation

## Problem Table

| #  | Problem                                   | Difficulty | Pattern                |
|----|-------------------------------------------|------------|------------------------|
| 65 | Print each character of string            | Easy       | Traversal              |
| 66 | Reverse a string                          | Easy       | Two Pointer            |
| 67 | Palindrome string (Two Pointer)           | Easy       | Two Pointer            |
| 68 | Toggle each alphabet                      | Medium     | Character Manipulation |
| 69 | Count strings with given prefix           | Medium     | Prefix Matching        |
| 70 | Capitalize first & last char of each word | Medium     | String Manipulation    |
| 71 | Maximum number of words in sentences      | Easy       | String Parsing         |
| 72 | Frequency of each character               | Medium     | Hashing                |
| 73 | Check anagram strings                     | Medium     | Hashing / Sorting      |
| 74 | Sort words in a sentence                  | Medium     | Sorting + Parsing      |

# COLLECTION FRAMEWORK

## Concepts Covered

- Set vs Map
- JavaScript Set, Map
- Dynamic data structures
- Hashing basics

## Problem Table

| #  | Problem                        | Difficulty | Pattern                   |
|----|--------------------------------|------------|---------------------------|
| 75 | Jewels and Stones              | Easy       | HashSet                   |
| 76 | Check if sentence is pangram   | Easy       | HashSet                   |
| 77 | First letter to appear twice   | Easy       | Hashing                   |
| 78 | Happy Number                   | Easy       | Hashing / Cycle Detection |
| 79 | Frequency of elements in array | Easy       | HashMap (Hashing)         |
| 80 | Two Sum                        | Easy       | Hashing                   |
| 81 | Sum of unique elements         | Easy       | HashSet + Traversal       |

| #  | Problem                    | Difficulty | Pattern             |
|----|----------------------------|------------|---------------------|
| 82 | Sort the People            | Medium     | HashMap + Sorting   |
| 83 | Most Frequent Even Element | Medium     | HashMap + Frequency |

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## Data Structures and Algorithmic Foundations

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# MEMORY AREAS & RECURSION – I

## Concepts Covered

- Stack memory, stack frames
- Recursion basics
- Base case & recursive calls

## Problem Table

| #  | Problem                   | Difficulty | Pattern                         |
|----|---------------------------|------------|---------------------------------|
| 84 | Print name N times        | Easy       | Recursion (Linear)              |
| 85 | Print 1 to N              | Easy       | Recursion (Forward)             |
| 86 | Print N to 1              | Easy       | Recursion (Backward)            |
| 87 | Sum of first N numbers    | Easy       | Recursion (Accumulation)        |
| 88 | Factorial of a number     | Easy       | Recursion (Multiplicative)      |
| 89 | Sum of digits (recursive) | Easy       | Digit Extraction + Accumulation |
| 90 | Reverse of a digit        | Easy       | Digit Manipulation              |
| 91 | Fibonacci number          | Easy       | Recursion (Tree)                |

# MATHEMATICS FOR PROGRAMMING

## Concepts Covered

- GCD, LCM
- Prime numbers, factors
- Logarithmic operations
- Sieve technique
- Fast exponentiation

## Problem Table

| #  | Problem                                  | Difficulty | Pattern             |
|----|------------------------------------------|------------|---------------------|
| 92 | GCD / HCF of two numbers                 | Easy       | Euclidean Algorithm |
| 93 | Find the Greatest Common Divisor (array) | Medium     | GCD Reduction       |
| 94 | Count primes (Sieve of Eratosthenes)     | Medium     | Sieve Technique     |
| 95 | Square root – Integer                    | Medium     | Binary Search       |
| 96 | Power ( $x^n$ )                          | Medium     | Fast Exponentiation |

## ARRAYS (LEVEL – 2)

### Concepts Covered

- Two Pointer
- Prefix Sum
- Sliding Window
- Greedy / Kadane's

### Problem Table

| #   | Problem                             | Difficulty | Pattern               |
|-----|-------------------------------------|------------|-----------------------|
| 97  | Merge Sorted Array                  | Easy       | Two Pointer           |
| 98  | Remove Duplicates from Sorted Array | Easy       | Two Pointer           |
| 99  | Duplicate Zeros                     | Easy       | Simulation            |
| 100 | Move Zeroes                         | Easy       | Two Pointer           |
| 101 | Sort Colors (Dutch National Flag)   | Medium     | Three Pointer         |
| 102 | Maximum Subarray (Kadane's Algo)    | Medium     | Kadane                |
| 103 | Majority Element (Boyer Moore)      | Medium     | Voting Algorithm      |
| 104 | Best Time to Buy & Sell Stock       | Easy       | Kadane / Greedy       |
| 105 | Trapping Rain Water                 | Hard       | Two Pointer           |
| 106 | Container With Most Water           | Medium     | Two Pointer           |
| 107 | 3Sum                                | Medium     | Sorting + Two Pointer |
| 108 | Merge Intervals                     | Medium     | Greedy + Sorting      |

## SECTION: SORTING ALGORITHMS

### Concepts Covered

- Merge Sort
- Quick Sort
- Cyclic Sort

## Problem Table

| #   | Problem                                  | Difficulty | Pattern     |
|-----|------------------------------------------|------------|-------------|
| 109 | Missing Number                           | Easy       | Cyclic Sort |
| 110 | Find All Numbers Disappeared in an Array | Easy       | Cyclic Sort |
| 111 | First Missing Positive                   | Hard       | Cyclic Sort |

## BINARY SEARCH

### Concepts Covered

- Binary Search (Iterative, Recursive)
- Search space reduction
- Binary Search on Answer

## Problem Table

| #   | Problem                                 | Difficulty | Pattern                 |
|-----|-----------------------------------------|------------|-------------------------|
| 112 | Search Insert Position                  | Easy       | Binary Search           |
| 113 | First and Last Position in Sorted Array | Medium     | Binary Search           |
| 114 | Count of element in sorted array        | Medium     | Binary Search           |
| 115 | Peak in Mountain / Bitonic array        | Medium     | Binary Search           |
| 116 | Search in Rotated Sorted Array          | Medium     | Binary Search           |
| 117 | Book Allocation Problem                 | Hard       | Binary Search on Answer |
| 118 | Capacity to Ship Packages               | Hard       | Binary Search on Answer |
| 119 | Koko Eating Bananas                     | Hard       | Binary Search on Answer |

## HASHING (LEVEL – 2)

### Concepts Covered

- Hashing fundamentals
- Collision handling
- HashMap / HashSet usage

## Problem Table

| #   | Problem                           | Difficulty | Pattern                  |
|-----|-----------------------------------|------------|--------------------------|
| 120 | Intersection of Two Arrays        | Easy       | HashSet                  |
| 121 | Subarray Sum Equals K             | Medium     | Prefix Sum + HashMap     |
| 122 | Longest Subarray with Sum K       | Medium     | Prefix Sum + HashMap     |
| 123 | Contiguous Array                  | Medium     | Prefix Sum + HashMap     |
| 124 | Longest Consecutive Sequence      | Medium     | HashSet                  |
| 125 | Count distinct elements in window | Medium     | Sliding Window + HashMap |

## OOPS INTRODUCTION

### Concepts Covered

- Class, Object
- Constructor, constructor function
- this keyword
- Prototype object
- Encapsulation (basics)
- Memory Efficiency – biggest advantage

### Problem Table

| #   | Problem                     | Difficulty | Pattern         |
|-----|-----------------------------|------------|-----------------|
| 126 | Design Private Cab Service  | Medium     | Object Modeling |
| 127 | Design Railway Ticket Class | Easy       | Class Design    |

## LINKED LIST

### Concepts Covered

- Linked List implementation
- Dynamic memory allocation
- Types of Linked List
- Two Pointer technique

### Problem Table

| #   | Problem            | Difficulty | Pattern        |
|-----|--------------------|------------|----------------|
| 128 | Design Linked List | Easy       | Implementation |

| #   | Problem                            | Difficulty | Pattern                       |
|-----|------------------------------------|------------|-------------------------------|
| 129 | Middle of Linked List              | Easy       | Two Pointer (Tortoise & Hare) |
| 130 | Merge Two Sorted Lists             | Easy       | Two Pointer                   |
| 131 | Reverse Linked List                | Easy       | Iterative / Recursion         |
| 132 | Linked List Cycle                  | Easy       | Cycle Detection (Floyd's)     |
| 133 | Remove Duplicates from Sorted List | Easy       | Traversal                     |
| 134 | Reverse Nodes in K Group           | Hard       | Linked List Manipulation      |
| 135 | Add Two Numbers                    | Medium     | Simulation                    |

## SECTION: STACK & QUEUE

### Concepts Covered

- Stack, Queue, Deque
- Implementation using arrays
- Expression evaluation
- Monotonic stack

### Problem Table

| #   | Problem                        | Difficulty | Pattern            |
|-----|--------------------------------|------------|--------------------|
| 136 | Implement Queue using Stacks   | Easy       | Simulation         |
| 137 | Implement Stack using Queues   | Easy       | Simulation         |
| 138 | Valid Parentheses              | Easy       | Stack (Validation) |
| 139 | Next Larger Element            | Easy       | Monotonic Stack    |
| 140 | Stock Span Problem             | Medium     | Monotonic Stack    |
| 141 | Largest Rectangle in Histogram | Hard       | Monotonic Stack    |
| 142 | Maximal Rectangle              | Hard       | Stack + Matrix     |

## NUMBER SYSTEM

### Concepts Covered

- Bitwise operators
- Bit manipulation techniques

### Problem Table

| #   | Problem                       | Difficulty | Pattern          |
|-----|-------------------------------|------------|------------------|
| 143 | Swap two integers (XOR)       | Easy       | XOR Trick        |
| 144 | Check even or odd (Bit)       | Easy       | Bit Check        |
| 145 | Check if i-th bit is set      | Easy       | Bit Masking      |
| 146 | Flip a specific bit           | Easy       | Bit Manipulation |
| 147 | Check if number is power of 2 | Easy       | Bit Trick        |

## RECURSION & BACKTRACKING – ADVANCED

### Concepts Covered

- Tail & Head Recursion
- Backtracking
- Recursion tree

### Problem Table

| #   | Problem                             | Difficulty | Pattern                |
|-----|-------------------------------------|------------|------------------------|
| 148 | Sort List using Recursion           | Medium     | Recursion              |
| 149 | Tower of Hanoi                      | Medium     | Recursion              |
| 150 | Josephus Problem                    | Medium     | Recursion              |
| 151 | Find Winner of Circular Game        | Medium     | Simulation / Recursion |
| 152 | Power Set                           | Medium     | Backtracking           |
| 153 | Subsets                             | Medium     | Backtracking           |
| 154 | Permutations                        | Medium     | Backtracking           |
| 155 | Subsets II                          | Medium     | Backtracking           |
| 156 | Letter Combinations of Phone Number | Medium     | Backtracking           |
| 157 | Combination Sum                     | Medium     | Backtracking           |
| 158 | Sudoku Solver                       | Hard       | Backtracking           |
| 159 | N-Queens                            | Hard       | Backtracking           |
| 160 | Word Search                         | Medium     | Backtracking           |

## BINARY TREE

### Concepts Covered

- Tree basics & terminology
- Traversals (DFS, BFS)
- Recursive tree processing

## Problem Table

| #   | Problem                             | Difficulty | Pattern          |
|-----|-------------------------------------|------------|------------------|
| 161 | Inorder Traversal                   | Easy       | DFS              |
| 162 | Preorder Traversal                  | Easy       | DFS              |
| 163 | Postorder Traversal                 | Easy       | DFS              |
| 164 | Maximum Depth of Binary Tree        | Easy       | DFS              |
| 165 | Level Order Traversal               | Easy       | BFS              |
| 166 | Symmetric Tree                      | Easy       | DFS              |
| 167 | Path Sum                            | Easy       | DFS              |
| 168 | Invert Binary Tree                  | Easy       | DFS              |
| 169 | Diameter of Binary Tree             | Medium     | DFS              |
| 170 | Same Tree                           | Easy       | DFS              |
| 171 | Left View of Binary Tree            | Medium     | BFS/DFS          |
| 172 | Top View of Binary Tree             | Medium     | BFS + Map        |
| 173 | Lowest Common Ancestor              | Medium     | DFS              |
| 174 | Path Sum III                        | Medium     | Prefix Sum + DFS |
| 175 | Construct Tree (Preorder + Inorder) | Medium     | Recursion        |

## BINARY SEARCH TREE (BST)

### Concepts Covered

- BST properties
- Search, insert, delete
- Inorder traversal (sorted property)

## Problem Table

| #   | Problem                     | Difficulty | Pattern            |
|-----|-----------------------------|------------|--------------------|
| 176 | Validate Binary Search Tree | Medium     | DFS (Bounds Check) |
| 177 | Kth Smallest in BST         | Medium     | Inorder Traversal  |
| 178 | Delete Node in BST          | Medium     | BST Manipulation   |

| #   | Problem                       | Difficulty | Pattern      |
|-----|-------------------------------|------------|--------------|
| 179 | Lowest Common Ancestor in BST | Easy       | BST Property |

## BINARY HEAP

### Concepts Covered

- Min Heap, Max Heap
- Heapify, HeapSort
- PriorityQueue
- Comparator logic

### Problem Table

| #   | Problem                      | Difficulty | Pattern                 |
|-----|------------------------------|------------|-------------------------|
| 180 | Last Stone Weight            | Easy       | Max Heap                |
| 181 | Kth Largest Element in Array | Medium     | Heap                    |
| 182 | Top K Frequent Elements      | Medium     | Heap + Hashing          |
| 183 | Top K Frequent Words         | Medium     | Heap + Sorting          |
| 184 | K Closest Points to Origin   | Medium     | Heap                    |
| 185 | Merge K Sorted Lists         | Hard       | Heap / Divide & Conquer |

## SLIDING WINDOW TECHNIQUE

### Concepts Covered

- Fixed & variable window
- Two Pointer (expand/shrink)
- Window optimization

### Problem Table

| #   | Problem                                   | Difficulty | Pattern                  |
|-----|-------------------------------------------|------------|--------------------------|
| 186 | Max Sum Subarray of size K                | Easy       | Fixed Window             |
| 187 | Count Occurrence of Anagram               | Medium     | Sliding Window + Hashing |
| 188 | Largest Subarray with Sum K               | Medium     | Variable Window          |
| 189 | Longest Substring Without Repeating Chars | Medium     | Sliding Window + Hashing |
| 190 | Maximum Points You Can Obtain             | Medium     | Sliding Window           |

| #   | Problem                  | Difficulty | Pattern                  |
|-----|--------------------------|------------|--------------------------|
| 191 | Minimum Window Substring | Hard       | Sliding Window + Hashing |
| 192 | Sliding Window Maximum   | Hard       | Deque / Monotonic        |

## GRAPH

### Concepts Covered

- BFS, DFS
- Cycle detection
- Topological sort
- Shortest path & MST

### Problem Table

| #   | Problem                         | Difficulty | Pattern          |
|-----|---------------------------------|------------|------------------|
| 193 | BFS Traversal                   | Easy       | BFS              |
| 194 | DFS Traversal                   | Easy       | DFS              |
| 195 | Cycle Detection (Undirected)    | Medium     | BFS/DFS          |
| 196 | Cycle Detection (Directed)      | Medium     | DFS              |
| 197 | Topological Sort (DFS)          | Medium     | DFS              |
| 198 | Topological Sort (Kahn's)       | Medium     | BFS              |
| 199 | Cycle Detection (Kahn's)        | Medium     | BFS              |
| 200 | Keys and Rooms                  | Medium     | DFS              |
| 201 | All Paths from Source to Target | Medium     | DFS              |
| 202 | Number of Islands               | Medium     | DFS/BFS          |
| 203 | Dijkstra Algorithm              | Medium     | Shortest Path    |
| 204 | Bellman Ford Algorithm          | Medium     | Shortest Path    |
| 205 | Prim's Algorithm                | Medium     | MST              |
| 206 | Kruskal Algorithm               | Medium     | MST              |
| 207 | Flood Fill                      | Easy       | BFS/DFS          |
| 208 | Clone Graph                     | Medium     | DFS/BFS          |
| 209 | Number of Provinces             | Medium     | DFS/BFS          |
| 210 | Course Schedule                 | Medium     | Topological Sort |
| 211 | Cheapest Flights Within K Stops | Hard       | BFS / Dijkstra   |

# DYNAMIC PROGRAMMING (DP)

## Concepts Covered

- Memoization, Tabulation
- Knapsack patterns
- String DP
- Partition DP

## Problem Table

| #   | Problem                                 | Difficulty | Pattern                 |
|-----|-----------------------------------------|------------|-------------------------|
| 212 | Fibonacci Number                        | Easy       | DP (1D)                 |
| 213 | Climbing Stairs                         | Easy       | DP (1D)                 |
| 214 | Min Cost Climbing Stairs                | Easy       | DP (1D)                 |
| 215 | Jump Game                               | Medium     | Greedy / DP             |
| 216 | 0/1 Knapsack                            | Medium     | DP (Knapsack)           |
| 217 | Subset Sum Problem                      | Medium     | DP (Subset)             |
| 218 | Partition Equal Subset Sum              | Medium     | DP (Subset)             |
| 219 | Perfect Sum Problem                     | Medium     | DP (Count Subsets)      |
| 220 | Find min difference between two subsets | Medium     | DP (Subset)             |
| 221 | Target Sum                              | Medium     | DP (Count Subsets)      |
| 222 | Unbounded Knapsack                      | Medium     | DP (Knapsack)           |
| 223 | Rod Cutting                             | Medium     | DP (Unbounded Knapsack) |
| 224 | Coin Change II                          | Medium     | DP (Count Ways)         |
| 225 | Coin Change                             | Medium     | DP (Min Coins)          |
| 226 | Longest Common Subsequence              | Medium     | DP (Strings)            |
| 227 | Longest Common Substring                | Medium     | DP (Strings)            |
| 228 | Print Longest Common Subsequence        | Medium     | DP (Reconstruction)     |
| 229 | Shortest Common Supersequence           | Medium     | DP (Strings)            |
| 230 | Longest Palindromic Subsequence         | Medium     | DP (Strings)            |
| 231 | Minimum Deletions to Make Palindrome    | Medium     | DP (Strings)            |
| 232 | Matrix Chain Multiplication             | Hard       | DP (Partition)          |
| 233 | Palindrome Partitioning                 | Hard       | DP (Partition)          |
| 234 | Longest Palindromic Substring           | Medium     | DP / Expand             |
| 235 | Delete Operation for Two Strings        | Medium     | DP (Strings)            |

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| #   | Problem             | Difficulty | Pattern        |
|-----|---------------------|------------|----------------|
| 236 | Egg Dropping Puzzle | Hard       | DP (Partition) |